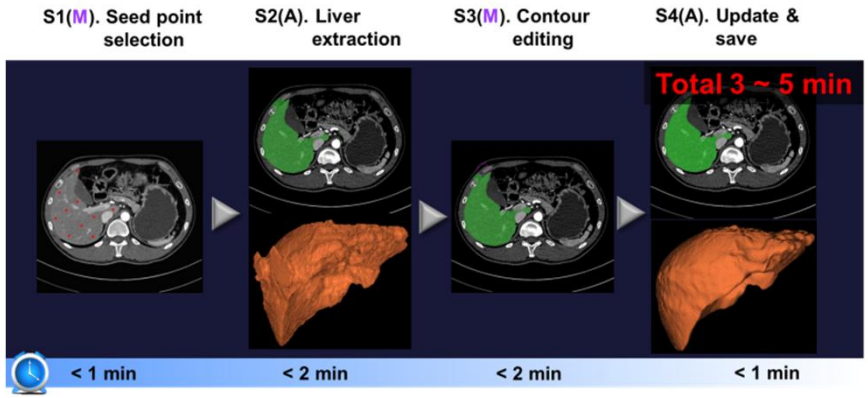
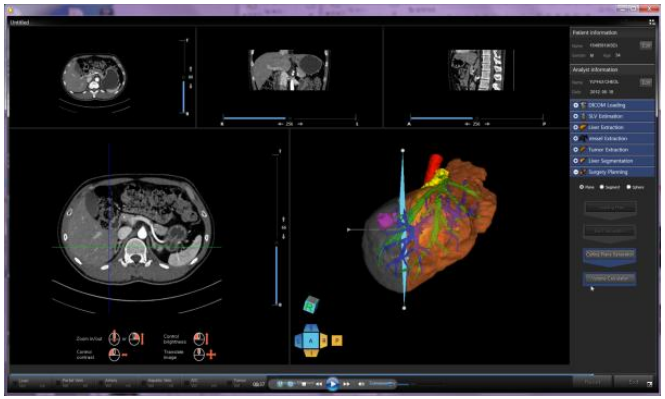
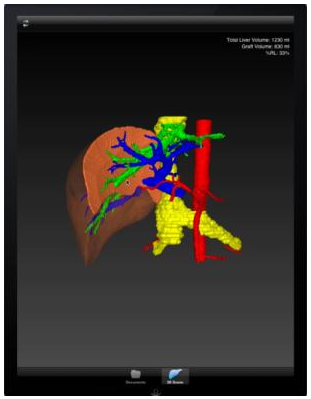
Information Input & Processing







Virtual liver surgery planning system (You et al., Jan. 2009 ~ present; sponsored by Ministry of Health & Welfare & Humanopia Co., Ltd.)

Contents

Part I

- Information Theory
- Information Display & Modality Selection
- Information Coding & Judgment
- Signal Detection Theory (SDT)

Part II

- Human Memory Systems
- Attention & Vigilance
- Compatibility & Aging
- Mental Workload & AI
- Quiz

Learning Goals (1/2)

GOALS

- Describe the **Sensory Storage, Working Memory, and Long-Term Memory** in the human memory system and explain **the role of attention** in transferring information between these stages.
- Define **chunking** as a mechanism to overcome Working Memory capacity limits and apply it to practical UI formatting.
- Explain the role of semantic networks and schemas in Long-Term Memory, and apply concepts like recognition over recall and elaborative rehearsal to improve user interface design.
- Differentiate between **Selective, Focused, Divided, and Sustained attention** and recognize the impact of load stress and distraction in monitoring tasks.
- Identify the causes of vigilance decrement and apply countermeasures to maintain operator alertness.

- Explain Wickens' **Multiple Resource Theory** using the dimensions of **Input Modalities, Processing Stages, Processing Codes, and Response Modalities** to predict multitasking interference.
- Define **Conceptual, Movement, Spatial, and Modality compatibility** and explain how high compatibility reduces mental recoding and improves response time.

Learning Goals (2/2)

GOALS

- Identify **age-related cognitive changes** and implement design modifications to reduce the cognitive burden for older users.
- Explain **mental workload** and identify **methods for mental workload assessment**.
- Assess multidimensional subjective workload utilizing the **six weighted subscales of the NASA-TLX**.
- Address challenges in **human-AI interaction** by focusing on the **knowledge acquisition** bottleneck, **system explainability**, and the **calibration of user trust**.

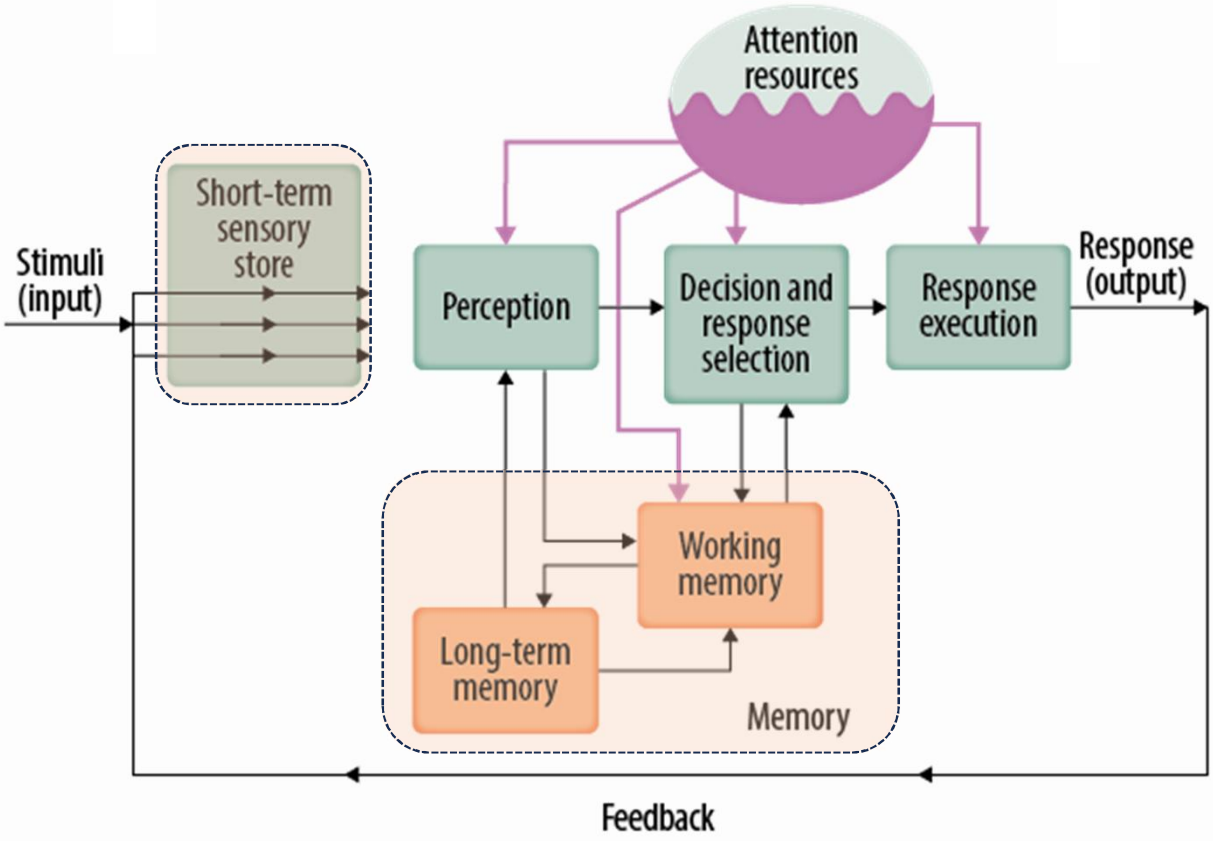
Section 05

Human Memory Systems

From sensory traces to working memory to long-term storage

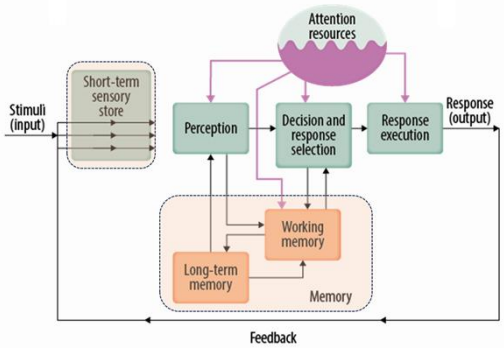
Human Information Processing (HIP) Model

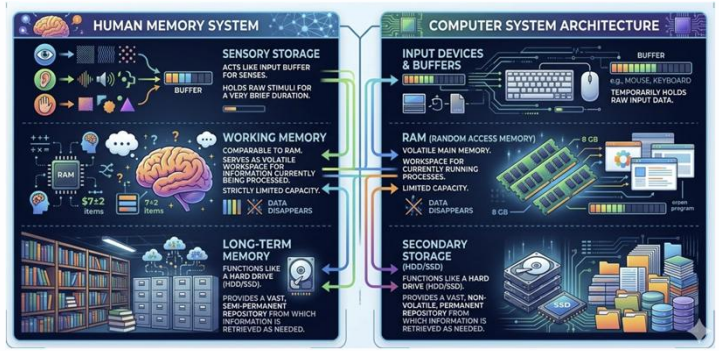
The Human Information Processing (HIP) model is a cognitive framework, describing how **stimuli is processed through sensory, working (short-term), and long-term memory systems to guide decisions and actions**.



Memory Architecture

The human memory architecture is remarkably similar to a computer's architecture





Human Memory Subsystem	Computer System	Duration	Capacity	Encoding	Key Function
Sensory Storage (Iconic/Echoic)	Input buffer for a mouse or keyboard	< 1 s (iconic) 2–4 s (echoic)	Very large (raw, unprocessed)	Physical/ Pre-attentive	Briefly prolongs stimulus for attention selection
Working Memory (Short-Term)	RAM	15–30 s (without rehearsal)	7 ± 2 chunks (Miller, 1956)	Visual, Phonetic, Semantic	Active manipulation and temporary storage of attended information
Long-Term Memory	HDD/SDD	Potentially permanent	Virtually unlimited	Semantic (primarily)	Permanent knowledge store; retrieval difficulty depends on organization

Sensory Storage: Iconic and Echoic

Iconic Storage (Visual)

A high-capacity but rapidly decaying visual persistence.

- Duration: Very brief (< 1 second)
- Function: Supports rapid scanning and stitching of visual scenes (saccadic integration)

Echoic Storage (Auditory)

Auditory traces that persist longer to aid language processing.

- Duration: Longer persistence (~2-4 seconds)
- Function: Essential for speech comprehension and integrating temporal patterns

Design Implications

- **Visual Transients:** Brief visual alerts must persist long enough or leave a "history" trace, as iconic memory fades too fast for complex analysis.
- **Auditory Alerts:** Use sound for temporal warnings; echoic memory allows users to "replay" the last few seconds of sound (e.g., "What did you say?").

Comparison

Feature	Iconic	Echoic
Modality	Visual	Auditory
Duration	< 1 sec	2-4 sec
Capacity	High (Snapshot)	Moderate

Working Memory: Capacity & Chunking

The "Magical Number"

Classic finding by Miller (1956) suggesting WM holds **7 ± 2 chunks** of information.

(Note) Recent research suggests realistic limit is often closer to 4 chunks in complex tasks.

What is a Chunk?

A unit of information defined by **meaningfulness** and **familiarity** to the user, not physical size.

Chocolate chunks

Capacity Constraints & Implications

Factor	Constraint / Characteristic
Limit	Pure capacity is small (limited resource pool). Risk: Overflow leads to errors/forgetting.
Duration	Requires active maintenance (rehearsal). Without rehearsal: Decays in ~10-20 sec.
Unit Type	Measured in **Chunks** (meaningful groups). "IBM" = 1 chunk; "I, B, M" = 3 chunks.
Design Fix	**Externalize Memory:** Don't make users hold information. Use visuals, menus, and breadcrumbs.

(Note) **Breadcrumbs**: **Small, and structured notes or cues left for oneself at the end of a work session** to easily resume the task later.
(e.g., website trail: Home > Furniture > Chairs > Leather Chairs)

[Chunking] Definition & Mechanism

Core Concept

Chunking is the process of **grouping individual pieces of information into familiar, manageable units** ("chunks").

Key Insight

Working memory limits are based on the *number of chunks*, not the amount of information within each chunk.

Effectively expands WM capacity

Mechanism Breakdown

Dimension	Explanation & Implication
Definition	Recoding raw stimuli into meaningful groups based on prior knowledge.
Mechanism	Relies on **Long-Term Memory (LTM)** associations to bind items together. Example: Seeing "F-B-I" as one unit (Agency) vs three letters.
Capacity Expansion	Allows holding more *total bits* of info within the 7±2 limit. ✓ Critical for expert performance.
Knowledge Dependency	Chunks must be **familiar** to the user to be effective. ⚠ Expert chunks may not work for novices.

[Chunking] Practical Examples in UI

UI Formatting Comparison

Data Type	Raw Input (Hard to Process)		Chunked UI (Easy to Process)
Credit Card	4444555566667777 16 items (exceeds 7±2)	→	4444 5555 6666 7777 4 manageable chunks
Phone Number	1234567890 10 items (borderline/hard)	→	(123) 456-7890 3 familiar chunks
Acronyms	I B M J F K T V 8 unrelated letters	→	IBM JFK TV 3 meaningful concepts
Menu Items	List of 15 mixed items High search cost	→	3 Categories of 5 items Hierarchical grouping

Cognitive Mechanism

Without Chunking

WM Overload!

↓

With Chunking

3 Units (Within Capacity)

Design Takeaway

Always format long strings of data (IDs, codes, dates) into groups of 3-4 characters. This simple UI change drastically reduces error rates and mental effort.

Working Memory: Codes

Coding Mechanisms

Visual Codes

Mental pictures and spatial layouts. Essential for navigation, maps, and graphical interfaces.

Example: Remembering a route on a map

Phonetic Codes

Auditory representations and inner speech. Used for language processing and retaining verbal instructions.

Example: Repeating a phone number internally

Semantic Codes

Abstract representations of meaning. Critical for comprehension and transfer to Long-Term Memory.

Example: Understanding the gist of a paragraph

Design Implications

Multimodal Coding

Combining codes (Visual + Phonetic) enhances memory retention through **Dual Coding Theory**. Information processed through multiple channels is more robust.

- ✓ Pair icons (visual) with labels (semantic/phonetic)
- ✓ Use spatial layout to reinforce logical groups

Interference Risk

Similar codes interfere with each other. Avoid requiring two simultaneous tasks using the same code.

✗ Reading + Listening (Both Verbal)	✓ Driving + Listening (Spatial + Verbal)

Working Memory: Rehearsal & Decay

Maintenance Rehearsal

Active repetition is required to keep information in Working Memory (WM).

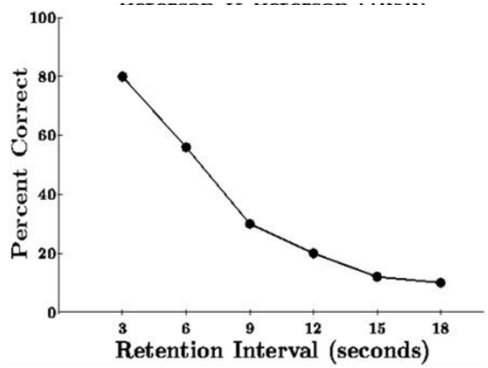
- Without rehearsal, information decays rapidly (within ~15-30 seconds).
- Rehearsal refreshes the memory trace, preventing loss.

Attention as a Gateway

Attention is the limited resource that enables rehearsal.

- Interference (distractions, new tasks) disrupts the rehearsal loop.
- Once interrupted, unrehearsed information is often permanently lost from WM.

Forgetting Curve (Peterson & Peterson, 1959)



Without rehearsal, recall drops to near zero in ~18 seconds.

Design Implications

- **Minimize Interruptions:** Do not interrupt users during tasks requiring WM storage (e.g., transcribing a code), as attention shifts break the rehearsal loop.
- **Enable Rapid Re-orientation:** If interruption is inevitable, provide visual placeholders or "breadcrumbs" to help users quickly resume their mental state.

Long-Term Memory: Retrieval

Semantic Networks & Schemas

Information is stored based on meaning and associations, not isolated facts.

- Knowledge is organized into schemas (mental frameworks) that aid retrieval.
- Stronger associations = faster and more reliable access.

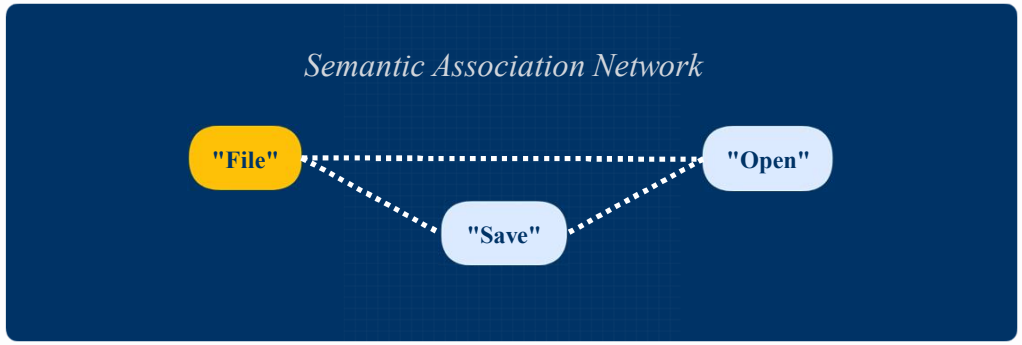
Recognition vs. Recall

Recognition is cognitively easier than recall because it provides cues.

- **Recall:** Retrieving info without cues (e.g., essay test).
- **Recognition:** Identifying info among options (e.g., multiple choice).

Design Guidelines

- **Promote Recognition:** Use menus and icons instead of command lines. "See and choose" is better than "remember and type."
- **Meaningful Categories:** Group items logically (e.g., File, Edit, View) to match user mental models (schemas).
- **Consistent Labeling:** Use standard terminology to trigger existing semantic associations.



Transfer from WM to LTM: Encoding

Working Memory

Limited Capacity
7 ± 2
Chunks

Active Processing
Temporary Hold

Semantic Encoding
Attaching meaning to new information

Organization
Structuring data hierarchically

Chunking
Grouping into familiar units

Encoding Process

Long-Term Memory

Unlimited Capacity
Durable Storage

Semantic Networks
Schemas & Scripts

Elaborative Rehearsal

Connecting new information to existing knowledge structures (schemas) deepens encoding and retention.

Mental Models

Organized structures that allow users to predict system behavior and store procedural knowledge efficiently.

Design Implication

Use **meaningful** headers, categories, and **consistent terminology** to support rapid transfer to LTM.

Design Guidelines

Key Principles

Effective interface design must respect the limitations of human working memory while leveraging the robust capacity of long-term memory structures.

- Goal: Reduce cognitive load
- Goal: Prevent error-prone reliance

Pro Tip

Always test your designs with users under time pressure to reveal true memory bottlenecks that might be hidden during relaxed testing.

Memory-Optimized Design Checklist

Use Meaningful Chunking

Group information into logical units of 3-4 items. Format long strings (e.g., credit cards, phone numbers) with spaces or delimiters.

Leverage Semantic Labels & Icons

Use intuitive icons and clear text labels that map directly to user's mental models (LTM), reducing the need for new learning.

Externalize Memory

Don't force users to remember system state. Use breadcrumbs, visible mode indicators, and "knowledge in the world" rather than in the head.

Minimize Concurrent Tasks

Avoid requiring time-sharing during critical encoding phases. Allow pauses for rehearsal if information retention is required.

Section 06

Attention & Vigilance

Selective, Focused, Divided, Sustained Attention; Vigilance Decrement & Multiple Resource Theory

Attention Taxonomy

What is Attention?

A limited resource pool that selects information for processing. It is not a unitary concept but consists of distinct varieties based on task demands.

Key Distinction

"Attention acts as a filter (selective) and a fuel (resource for divided tasks)."

Resource Implications

- **Narrow Focus:** High intensity, low breadth
- **Broad Focus:** Low intensity, high breadth
- **Failures:** Distraction, tunneling, fatigue

Four Varieties of Attention

Wickens et al. (2015)

Type	Definition & Goal	Real-World Example
Selective	**Monitoring multiple sources** Choosing the right channel to process; scanning.	Pilot scanning dashboard instruments to detect anomalies.
Focused	**Concentrating on one source** Excluding/ignoring distractions (inhibition).	Surgeon focusing on incision, ignoring monitor alarms.
Divided	**Parallel processing (Time-sharing)** Performing 2+ tasks simultaneously.	Driving while talking to a passenger or navigating.
Sustained	**Long-term monitoring (Vigilance)** Detecting infrequent signals over time.	TSA agent scanning luggage for 30+ minutes.

Selective vs. Focused Attention

Selective Attention

Definition

Monitoring multiple sources (channels) of information to perform a single task. Requires scanning and sampling strategies.

Key Example

A pilot scanning various cockpit instruments (altitude, speed, heading) to maintain stable flight.

Cockpit Scanning

Design Trade-off

Challenge: Cognitive tunneling or missing critical updates in unmonitored channels.

Design Tip: Place frequently scanned displays close together (visual proximity).

Focused Attention

Definition

Concentrating on one source while actively excluding or inhibiting processing of other distracting sources.

Key Example

Reading a technical manual in a noisy factory environment while ignoring conversations and machine noise.

Reading in Noise

Design Trade-off

Challenge: Failure to focus due to distraction (interference) or hyper-focus missing alarms.

Design Tip: Separate competing channels (e.g., distinct spatial locations or modalities).

Divided Attention: Time-Sharing

Dual-Task Performance

Performance depends on shared resources from a limited pool.

- Resource Competition: Tasks competing for the same resource (e.g., visual) degrade performance more than tasks using different resources.
- Scheduling: Ideally, operators switch attention rapidly rather than truly processing in parallel.

Automaticity

With extensive practice, tasks become automatic and demand fewer attentional resources.

- Example: Experienced drivers can converse while driving; novices cannot.

Factors Affecting Time-Sharing

Task Difficulty, Practice, and Structural Similarity (e.g., both visual).

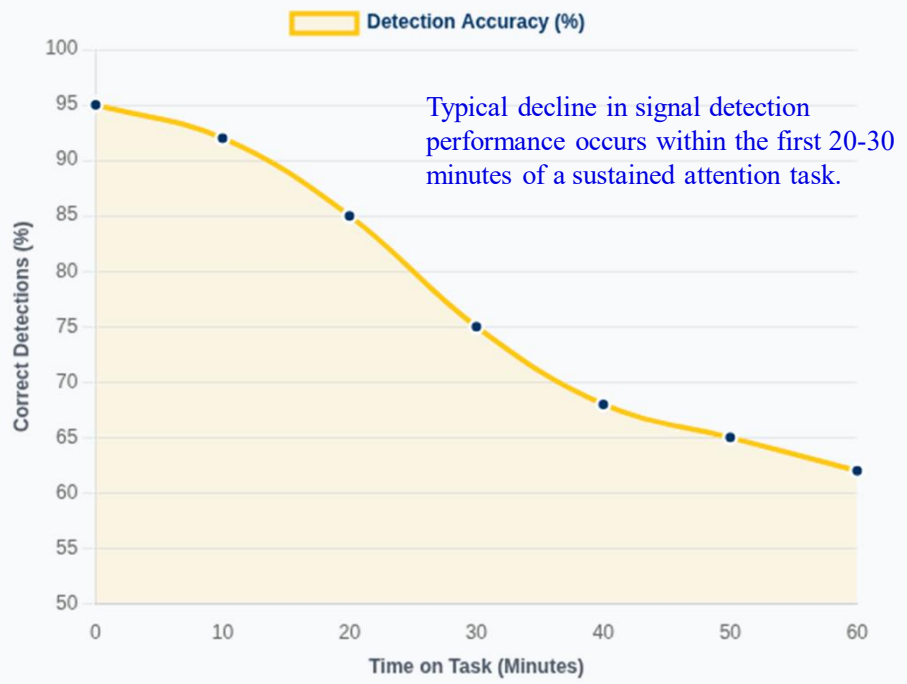
Resource Allocation & Performance

Design Implications

- **Minimize Concurrent Demands:** Avoid requiring simultaneous processing of high-load tasks. Stagger demands if possible.
- **Avoid Structural Overlap:** Don't require two visual tasks or two manual responses at the exact same time (see MRT).
- **Promote Automaticity:** Standardize interfaces to allow operators to develop automatic processing for routine tasks.

Sustained Attention: Vigilance Decrement

Performance Over Time: Vigilance Decrement



Underlying Causes

Time-on-Task Fatigue

Continuous monitoring depletes cognitive resources, leading to a progressive loss of sensitivity (d').

Signal Probability & Uncertainty

Rare signals (low probability) are harder to detect. Spatial and temporal uncertainty further exacerbates the decrement.

Design Implications

- **Increase Conspicuity:** Make signals larger, brighter, or louder to combat sensitivity loss.
- **Artificial Signals:** Inject artificial signals (with feedback) to keep operators alert and calibrate their criteria.
- **Work-Rest Schedules:** Brief breaks can restore performance levels significantly.

Countermeasures for Vigilance

To mitigate the vigilance decrement (performance decline over time), apply these evidence-based interventions.

Wickens et al. (2015)

Work-Rest Schedules

Limit watch duration to 20-30 minutes before breaks. Rotation between vigilance and active tasks prevents fatigue accumulation.

Enhance Signal Conspicuity

Increase signal intensity, duration, or size. Make targets distinct from noise/background to reduce perceptual load.

Provide Feedback

Immediate knowledge of results (KR) improves detection. Inform operators when they miss a signal or make a false alarm.

Inject Artificial Signals

Insert "false" signals that require response. Increases event rate, maintains arousal, and allows for frequent feedback.

Reduce Uncertainty

Minimize spatial uncertainty (where) and temporal uncertainty (when). Use cuing to direct attention to likely signal locations.

Optimize Environment

Maintain comfortable temperature, lighting, and noise levels. Moderate noise (white noise) can sometimes aid alertness vs. total silence.

Adequate Training

Ensure operators know exactly what critical signals look like (template matching). Overlearning reduces cognitive effort.

Control Event Rate

Avoid very low event rates (under-arousal) or extremely high rates (overload). Pace signals if system allows.

Resource Theories: Single vs. Multiple

Single Resource Theory

One central "fuel tank" of capacity shared by all mental processes.

- Performance drops if total demand > available capacity
- Tasks interfere regardless of type (e.g., reading + listening)

Multiple Resource Theory

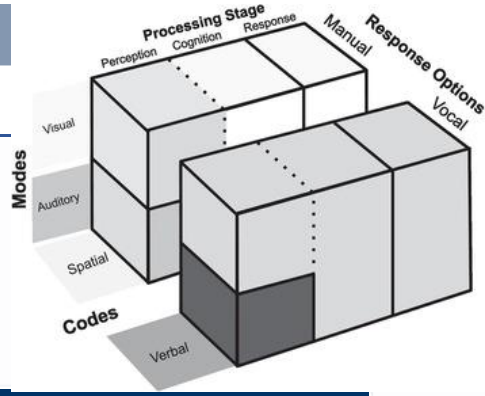
Independent pools of resources based on processing structures.

- Resources divided by: Modality, Code, Stage, Response
- Explains why we CAN drive (spatial) and talk (verbal) simultaneously

Design Implication

- **Separate Resources:** To enable effective time-sharing, design tasks to draw from different pools (e.g., Audio + Visual is better than Visual + Visual).
- **Avoid Overload:** Even with multiple pools, central executive control is limited. Don't saturate any single channel.

Wickens' Multiple Resource Theory



Wickens' Multiple Resource Theory (MRT) proposes **four categorical dimensions that define separate resource pools**. **Tasks sharing levels on these dimensions interfere more**.

Dimension	Description & Levels	Examples
Input Modalities	Visual vs. Auditory Seeing and hearing use different input channels.	Reading Map (Visual) vs Listening to Radio (Auditory) *Easier to combine than two visual tasks.*
Processing Stages	Perceptual/Cognitive vs. Response Resources for thinking vs. acting are separate.	Mental Arithmetic (Cognitive) vs Steering (Response) *Low interference between these stages.*
Processing Codes	Spatial vs. Verbal Spatial manipulation vs. language processing.	Mental Rotation (Spatial) vs Speech Recognition (Verbal) *Distinct cognitive codes minimize conflict.*
Response Modalities	Manual vs. Vocal Hand movements vs. speaking.	Typing/Steering (Manual) vs Talking (Vocal) *Can often be performed simultaneously.*

MRT Interference Matrix

Resource Conflict Logic

Interference between two tasks is predicted by the extent to which they compete for the same pool of resources along Wickens' four dimensions.

Conflict Equation

If (Task_A.Resources = Task_B.Resources)

Then Interference = **High**

Else If (Task_A.Resources ≠ Task_B.Resources; Dimensions differ)

Then Interference = **Low**

Key Insight

Performance decrement is greatest when tasks **overlap** in **Modality** (e.g., two visual tasks) or **Processing Code** (e.g., two verbal tasks).

Interference Prediction Table

	Task B: Visual/Spatial	Task B: Auditory/Verbal
Task A: Visual/Spatial	High Conflict	Low Conflict
Task A: Auditory/Verbal	Low Conflict	High Conflict

Example

Task 1	Manual Tracking (Visual + Spatial)
Task 2	Mental Math (Auditory + Verbal)
Result	**Effective Time-Sharing**

Case: Driving + Phone Conversation

Primary Task — Driving Analysis

Visual Channel
Road scanning, mirrors, dashboard

Processing Code
Spatial (lane keeping, gap estimation)

Response
Manual (steering, pedals)

"Driving is primarily a visual-spatial-manual task"

Scenario A: Hands-free Talking

Auditory Input | Verbal Code | Vocal Response

Interference Level: **Low (Different Resources)**

MRT Verdict:
Uses **different** modalities & codes than driving. Time-sharing is possible but central processing still competes.

Scenario B: Reading Text / Texting

Visual Input | Verbal Code | Manual Response

Interference Level: **High (Resource Conflict)**

MRT Verdict
Competes for **Visual** input and **Manual** response. Catastrophic interference expected.

Research Findings

- Texting increases crash risk by **23x** (VTTI).
- Even hands-free causes "cognitive tunneling" (Strayer et al.).

Design Implication

- Lock visual-manual tasks when vehicle is in motion.
- Use auditory alerts/voice command for navigation.

Design Guidelines

Resource Management

Successful multitasking design relies on minimizing competition for shared attentional resources across perceptual, cognitive, and response stages.

Goal: Distribute load across separate pools (Visual/Spatial vs. Auditory/Verbal)

Redundancy Caution

While redundant coding (audio+visual) helps detection, ensure it doesn't overload sensory channels during peak demand moments.

MRT-Based Design Principles

Separate Resource Pools

Design concurrent tasks to use different modalities (e.g., visual display + auditory alert) and codes (spatial map + verbal instruction) to minimize interference.

Minimize Visual Scanning

Co-locate related visual channels spatially to reduce scanning costs. Place critical indicators near the primary line of sight.

Use Auditory Channel for Alerts

Offload time-critical, omnidirectional warnings to the auditory channel, reserving visual bandwidth for complex spatial data. Beware of masking.

Avoid Concurrent Peak Demands

Stagger high-workload periods across tasks. Don't require intense visual search simultaneously with complex manual control.

Section 07

Compatibility & Aging

Aligning stimuli and responses with human expectations. Designing for older adults' cognitive processing.

What is Compatibility?

Definition

Compatibility refers to the **relationship of stimuli and responses** to **human expectations**. It is the degree to which relationships are consistent with human expectations.

Matching Expectations

Aligning system design with user's internal mental models.

- Reduces the need for "mental recoding"
- Leverages population stereotypes (e.g., up = on/increase)

Benefits of High Compatibility

- Faster learning times
- Faster reaction times
- Fewer errors, especially under stress
- Reduced mental workload

Four Dimensions of Compatibility

1. Conceptual

Code-meaning match (e.g., airplane icon for airport).

2. Movement

Control movement vs. system response (e.g., clockwise to increase).

3. Spatial

Physical arrangement of controls and displays (e.g., stove mapping).

4. Modality

Task-modality fit (e.g., verbal tasks with auditory input).

Insight: High compatibility removes the extra cognitive step of "translating" the stimulus before acting, leading to automatic processing.

Conceptual Compatibility

What is it?

Conceptual compatibility refers to the degree to which codes and symbols correspond to the **conceptual associations** (mental models) people already possess.

High	→	Low
Compatibility		Recoding Effort

Consistent Metaphors

Desktop "Trash Can"
Uses a physical world metaphor for deleting files. Users instantly understand "putting a file in the trash" deletes it but allows retrieval before emptying.

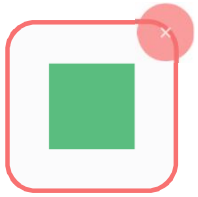
Folder Hierarchy
Mimics physical filing cabinets. "Documents inside folders" is a strong conceptual match for organizing digital data.

Design Rule

Always leverage existing **population stereotypes**. Do not force users to learn new arbitrary codes when a standard exists.

Iconography & Mapping

Airplane Icon
Matches "Airport"



Green Square
Abstract / No Meaning

Users process meaningful icons faster because they align with existing mental models (schemas).

Color Coding Standards

Danger / Halt

Warning / Yield

Safe / Proceed

Violating population stereotypes (e.g., Green = Stop) increases reaction time and error rates drastically.

Movement Compatibility

Clockwise for Increase

Standard convention for rotary controls (knobs, valves).

- Clockwise = Volume Up / Value Increase / Close Valve
- Counter-Clockwise = Volume Down / Value Decrease / Open Valve

Up/Right for Increase

Standard for linear controls (levers, sliders).

- Up/Forward/Right = Increase / On
- Down/Backward/Left = Decrease / Off

Control-Display Congruence

Control movement should match the direction of the display indicator movement (e.g., moving a lever up moves the pointer up).

Population Stereotypes

Expectations about control operation shared by a specific group of people. Violating these stereotypes increases reaction time and error rates, especially under stress.

Example: Light Switches

North America	Europe/Other
Up = On	**Down = On**

Movement Mapping Examples

Rotary Control

Clockwise to Increase

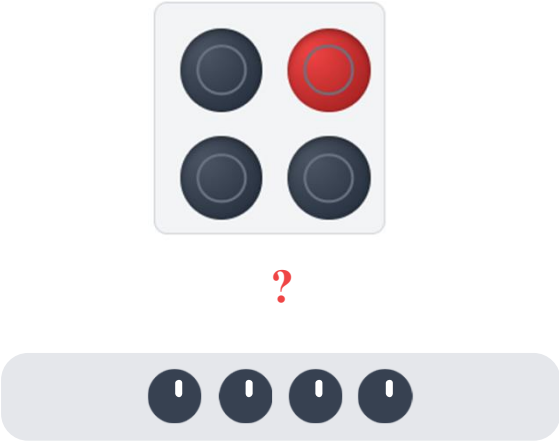
Linear Slider

Up to Increase

Spatial Compatibility

Low Compatibility

Design A: Linear Controls

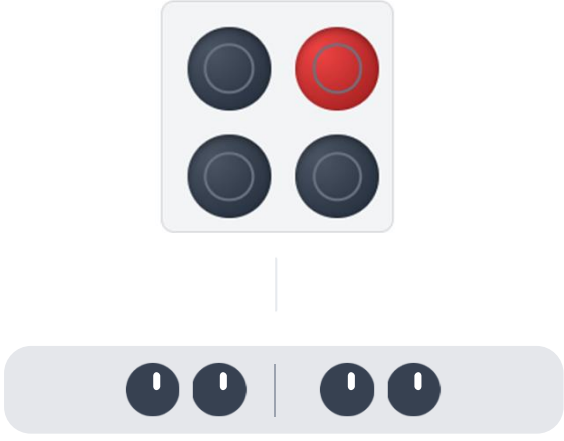


Problem

Which knob controls the back-right burner? Requires cognitive effort to map linear array to 2D grid.

Medium Compatibility

Design B: Grouped Offset

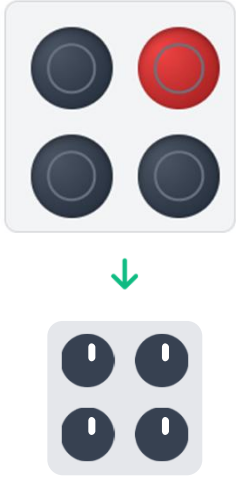


Improvement

Splitting controls into left/right groups helps, but specific burner assignment is still ambiguous.

High Compatibility

Design C: Spatial Isomorphism



Solution

Arranging controls in the same spatial pattern as the burners eliminates mapping ambiguity.

Core Principle: **Direct spatial mapping reduces mental transformation load**, leading to faster reaction times and fewer errors (especially under stress).

Modality Compatibility

Verbal Tasks

Language Processing

Optimal Modality

Auditory Presentation + Vocal Response

- Phonological loop handles auditory verbal information efficiently
- Spoken response avoids translation to manual codes

Real-World Application

Air Traffic Control

Complex verbal instructions delivered via radio headset are more compatible than reading text while scanning radar.

Spatial Tasks

Analog / Geometric

Optimal Modality

Visual Presentation + Manual Response

- Visuospatial sketchpad processes imagery and location directly
- Manual control (steering, pointing) maps naturally to space

Real-World Application

Drone Operation

Visual camera feed combined with joystick control offers superior performance over voice commands for navigation.

Design Guidelines: Compatibility

Core Concept

Maximize Stimulus-Response (S-R) Compatibility to reduce cognitive load, decrease reaction time, and minimize errors, especially under stress.

- Goal: Intuitive Interaction
- Goal: Reduce Training Time

Pro Tip

When full compatibility isn't possible, prioritize movement compatibility over spatial arrangement, as users rely heavily on learned motor stereotypes.

S-R Compatibility Guidelines

Use Natural Mappings

Design controls to physically resemble or spatially map to the system being controlled (e.g., seat adjustment controls shaped like a seat).

Group Controls Under Displays

Maintain spatial proximity and alignment. Place control knobs directly below or adjacent to their corresponding indicators to avoid cross-mapping errors.

Align Control Movement

Follow population stereotypes: Clockwise/Up/Right should generally mean "Increase", "On", or "Forward". Ensure feedback matches input direction.

Match Modality to Task Code

Use auditory/vocal modalities for verbal tasks (e.g., warnings, communication) and visual/manual modalities for spatial tasks (e.g., navigation, tracking).

Aging & Human Information Processing (HIP)

Onset of Effects

Age-related cognitive changes typically become noticeable after approximately 65 years of age.

- Progressive nature throughout adulthood
- Variability between individuals increases with age

General Slowing

A pervasive slowing in speed of processing across cognitive and motor tasks.

- Slower retrieval from Long-Term Memory
- Increased time for response selection and execution

Working Memory Vulnerability

Working Memory becomes significantly more fragile, especially during attention shifts.

- Increased disruption by interruptions or divided attention
- Difficulty holding information while processing other data

Performance Impact

Vocabulary
Remains Stable

Fluid Intelligence
Declines w/ Age

Note: Crystallized intelligence (knowledge/experience) is often preserved, while fluid intelligence (processing speed/WM) declines.

Age-Related Changes in HIP

The "Aging Effect"

Cognitive changes become noticeable after age ~65. While crystallized intelligence (experience) remains stable, fluid intelligence (processing speed, WM) declines.

Key Insight

Design must compensate for these deficits rather than rely on user adaptation.

Implications

- ✓ Simpler interfaces benefit all users
- ✓ Redundancy aids error recovery
- ✓ Consistent mapping is critical

Five Key Functional Changes

Change Category	Description & Impact
1. General Slowing	Slower retrieval from LTM, decision-making, and motor execution. Impact: Timed tasks become difficult.
2. WM Disruption	More easily disrupted by attention shifts; harder to "hold" info while processing. Impact: Multi-step navigation errors.
3. LTM Difficulty	Harder to search LTM and transfer new info from WM to LTM (encoding). Impact: Learning new systems takes longer.
4. Incompatibility	Greater difficulty dealing with unnatural mappings (e.g., up = off). Impact: Reverting to stereotypes under stress.
5. Perceptual Deficits	Trouble identifying objects from incomplete/ambiguous stimuli. Impact: Need clear, high-contrast signals.

Design Guidelines: for Older Users

Aging & Design

Modifications for older adults often benefit all users (Universal Design). The goal is to compensate for sensory and cognitive declines without stigmatizing the user.

- Sensory Compensation
- Cognitive Support
- Motor Accommodation

Design Insight

Simplicity is key. Reducing feature bloat helps older users maintain focus and reduces the burden on working memory.

Modifications for Cognitive Aging

Strengthen Signals (Sensory)

Increase contrast, font size, and audio volume. Use brighter, distinct icons to overcome reduced visual acuity and hearing loss.

Reduce Time-Sharing Demands

Design step-by-step sequential tasks rather than multitasking. Avoid requiring simultaneous monitoring of multiple displays.

Maximize Compatibility

Rely heavily on population stereotypes and natural mappings. Avoid abstract symbols that require new learning or mental rotation.

Provide Cues & Previews

Support "prospective memory" with clear reminders of what to do next. Show previews of actions before commitment.

Support Error Recovery

Make "Undo" prominent. Provide clear, non-threatening feedback when errors occur, guiding the user back to the correct path.

Section 08

Mental Workload & AI

Measuring mental workload & AI interfaces

Mental Workload

Core Concept

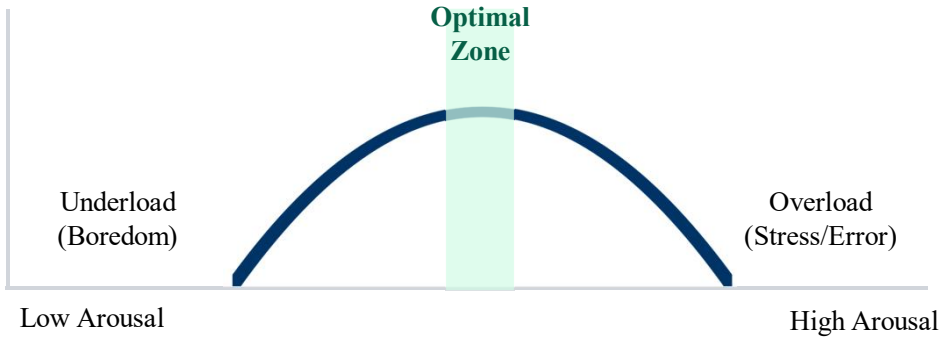
The relationship between the cognitive resources required by a task and the resources available to the operator.

Workload = Demand / Capacity

Influencing Factors

- Task Complexity & Structure
- Time Pressure & Rate
- Operator Skill & State (Fatigue/Stress)

Performance Curve



Design Implications

- **Manage Demand Peaks:** Flatten workload spikes by staggering tasks or automating routine functions during high-stress periods.
- **Resource Allocation:** Design interfaces that offload working memory (e.g., checklists, visual aids) to free up cognitive capacity.

Measurement Categories

Measuring Workload

Mental workload is a **multi-dimensional construct**. No single measure captures all aspects of processing demand.

Selection Strategy

Researchers often **use a battery of measures** combining these categories to triangulate actual cognitive cost.

Categories of Workload Measurement

Category	Description	Examples
Primary Task	Performance on the main task itself.	• Tracking error (RMS) • Response time • Accuracy / Error rate
Secondary Task	Performance on a concurrent, lower-priority task to gauge spare capacity.	• Time estimation • Mental arithmetic • Memory probe task
Physiological	Bodily responses reflecting arousal or cognitive effort.	• Heart rate variability (HRV) • Pupil diameter • P300 ERP (brain waves)
Subjective	User's self-reported perception of effort and difficulty.	• NASA-TLX • SWAT (Subj. Workload Assessment) • Modified Cooper-Harper

NASA-TLX

Six Subscales Structure

A multi-dimensional rating procedure that derives an overall workload score based on a weighted average of six dimensions.

- Mental Demand
- Physical Demand
- Temporal Demand
- Performance
- Effort
- Frustration

Weighting Procedure

Unique feature: Subscale ratings are weighted by their subjective importance to the specific task.

- Users perform pairwise comparisons (e.g., "Mental Demand vs. Effort") to determine weights.
- Result: A sensitive, individualized overall workload score (0-100).

Evaluation of NASA-TLX

Advantages

- Easy to administer and inexpensive
- Widely adopted standard (high comparability)
- Captures the "essence" of subjective workload

Limitations

- Post-hoc measure (memory bias possible)
- Can dissociate from actual performance (e.g., high effort to maintain performance)

When to Use

Best used when you need a global assessment of workload across different task types, and immediate, continuous measurement isn't required.

NASA-TLX: Six Subscales

The **NASA Task Load Index (NASA-TLX)** measures workload across six dimensions. Users rate each dimension on a scale (Low-High or Good-Poor) and weight them by importance.

Subscale Dimension	Definition	Example Question
Mental Demand Cognitive Load	How much mental and perceptual activity was required? (thinking, deciding, calculating, remembering, looking, searching, etc.)	*"Was the task easy or demanding, simple or complex?"*
Physical Demand Physical Exertion	How much physical activity was required? (pushing, pulling, turning, controlling, activating, etc.)	*"Was the task easy or demanding, slow or brisk, slack or strenuous?"*
Temporal Demand Time Pressure	How much time pressure did you feel due to the rate or pace at which the tasks or task elements occurred?	*"Was the pace slow and leisurely or rapid and frantic?"*
Performance Self-Assessment	How successful do you think you were in accomplishing the goals of the task set by the experimenter (or yourself)?	*"How satisfied were you with your performance?" (Good to Poor)*
Effort Total Work	How hard did you have to work (mentally and physically) to accomplish your level of performance?	*"How hard did you have to work to maintain your performance?"*
Frustration Stress / Annoyance	How insecure, discouraged, irritated, stressed and annoyed versus secure, gratified, content, relaxed and complacent did you feel?	*"Did you feel stressed and annoyed during the task?"*

NASA-TLX: Weighted Scoring

Calculation Steps

The weighted workload score accounts for individual differences in task perception by combining subjective importance weights with raw ratings.

1 Pairwise Weighting (W)

Compare 15 pairs of factors (e.g., Mental vs. Physical). Select the more relevant factor. Count selections for **each factor to get Weight (0-5)**. **Sum of weights = 15**.

2 Raw Rating (R)

Rate each of the **6 dimensions on a scale from 0 (Low/Good) to 100 (High/Poor)** based on the task experience.

3 Weighted Score Calculation

Multiply each Rating by its Weight (W × R). Sum these products and divide by 15.

Weighted Workload Equation

$$\text{Score} = \Sigma\,(\text{Weight}_i \times \text{Rating}_i) / 15$$

where i = each of the 6 subscales

Example

Participant A: Air Traffic Control

Subscale	Weight (W)	Rating (R)	Product (W×R)
Mental Demand	5	90	450
Physical Demand	0	20	0
Temporal Demand	4	80	320
Performance	3	50	150
Effort	2	70	140
Frustration	1	40	40

Sum of Products:	1,100
Total Weight Divisor:	/ 15
Overall Workload Score:	**73.3**

Choosing Workload Measures

Assessment Strategy

Selecting the appropriate mental workload measure requires balancing the need for data precision with the practical constraints of the operational environment.

- Goal: Valid & Reliable Data
- Avoid: User Interference

Trade-off Matrix

High Sensitivity Secondary Tasks	**Low Intrusion** Subjective (Post)
Continuous Physiological	**Practicality** Primary Task

Workload Measure Selection Criteria

Match Measure to Context & Constraints

Assess environmental limits first. Can sensors be worn? Can the primary task be interrupted? Does the task allow for a secondary probe?

Combine Multiple Measures (Triangulation)

Use a "converging operations" approach. Combine primary task performance data with subjective ratings (NASA-TLX) and physiological metrics where feasible.

Balance Sensitivity vs. Intrusiveness

Choose measures sensitive enough to detect relevant workload shifts without interfering with the primary task or adding artificial load.

Consider Diagnosticity

Select measures that reveal *why* workload is high (e.g., visual load vs. cognitive complexity) to inform specific design improvements.

AI Systems: Overview

Knowledge Capture

Programs designed to emulate human expert problem-solving abilities.

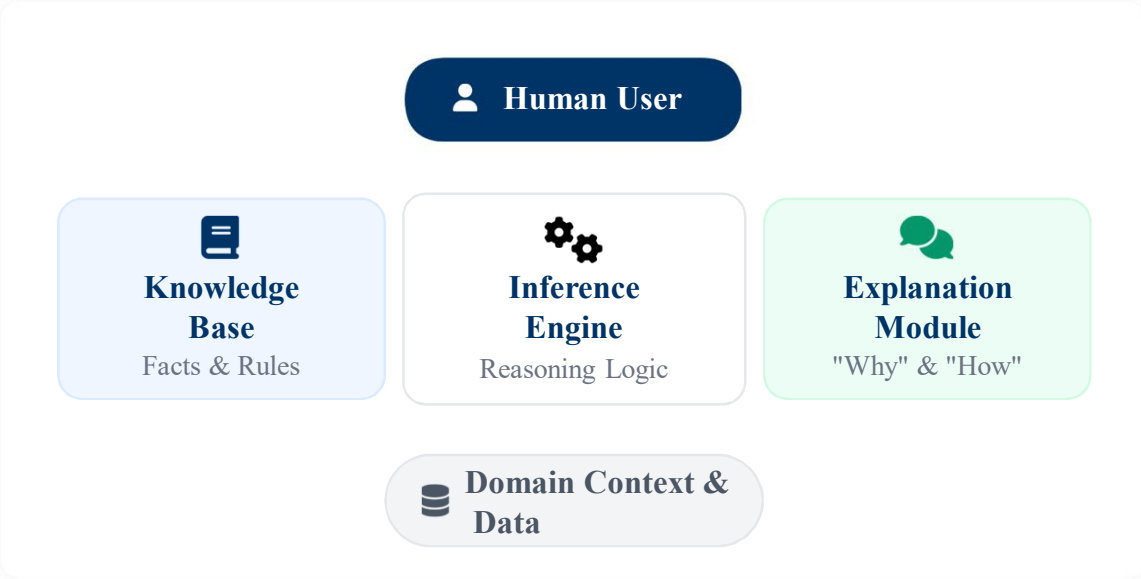
- Captures heuristic knowledge (rules of thumb)
- Encodes domain expertise into executable rules

Transparency Criticality

Systems must explain "why" a decision was made, not just "what".

- Builds user confidence and trust
- Allows human intervention for error correction

AI System Components



Key Challenge: The "Knowledge Acquisition Bottleneck" - **Extracting tacit knowledge from experts** is difficult and time-consuming.

AI Systems: Challenges from HF/E Aspects

Knowledge Elicitation

Expert's Tacit Knowledge

Intuition, experience, context

Translation Loss

IF *x* THEN y

Explicit Rules

"Experts know more than they can say and say more than they can write down."

The "Why" Problem (Explainability)

Critical

Systems must communicate not just **what** the decision is, but **why** it was reached.

Opaque System

"Recommendation: Shut down pump B."

Transparent System

"Shut down pump B because pressure > 400psi and valve C is stuck."

Trust & User Intervention

Safety

- **Over-trust (Complacency):** Users accept wrong advice without checking.
- **Under-trust (Disuse):** Users ignore correct advice after one failure.
- **Intervention Path:** Users must be able to redirect an obviously errant decision path.

Context Sensitivity

Machines struggle with ambiguity ("Washington" = city vs. state vs. person) that humans resolve instantly.